

Taiwan PCB/CCL: Compeq mgmt visit: Optical transceiver to be the key growth driver, with high ASP/volume/GM expansion potential; Buy ZDT,

We held a meeting with Compeq's (2313.TW, Not Covered) management. Key takeaways include:

(1) The company's current guidance is for revenue to reach NT\$110-120bn in 2027 (implying 23% 2025-27 CAGR from the midpoint, tracking faster than previous guidance of reaching NT\$100bn+ by 2028). The company believes that optical module PCB remains the key growth driver and expects the business revenue to reach NT\$25bn+ (7x+ growth vs. 2026 guidance) with increasing contribution from 1.6T optical module PCB with better pricing/margin (the company mentioned 800G/1.6T optical module PCB with GM at 40%+ vs. the company's GM at 18.6% in 2Q26). Also, management expects LEO-related business to reach NT\$25bn+ (30%+ YoY vs. 2026 guidance), with more contribution from high-end PCB in LEO satellites, driven by the ramp-up next gen LEO satellites from 4Q26/1Q27.

(2) For 2026, the company expects optical-related business to reach ~NT\$3bn (vs. very limited contribution in 2025), driven by the ramp-up of 800G optical module PCB from 3Q26, with ~NT2bn of the revenue expected to be recognized in 4Q26.

(3) The company expects LEO-related business revenue to reach ~NT\$19bn in 2026 (implying ~30% YoY vs. ~NT\$15bn in 2025), mainly driven by strong demand for PCB in ground terminal, and the company expects growth to outpace that of high-end PCB in LEO satellites.

(4) The company remains proactive with its capacity expansion plan and expects CAPEX to reach NT\$20bn+ in both 2026/27 (vs. ~NT\$7bn in 2025). The company will allocate 50% of its 2026 CAPEX to expand optical module PCB capacity, with the remaining 50% allocated to high-end PCB, primarily to support LEO-related business. The company mentioned its target to ramp optical module PCB capacity to reach monthly revenue of ~NT\$3bn by 2027 end (vs. less than NT\$100mn per month of optical-related business contribution as of 2Q26 vs. the target monthly revenue of 45% of Compeq's average monthly revenue YTD). Despite the significant capacity expansion, the company noted that it expects to fulfill only 60-70% of total customer demand in 2027.

(5) The company is seeing multiple growth drivers from the key LEO customer, and the customer is asking the company to expand its PCB capacity by 5x+ over the next 3-5 years to meet its strong demand. Per management the drivers include: (1) migration to next-gen LEO satellites, with PCB area consumption expected to increase 2x+ vs. the current gen, and expected to ramp-up from 4Q26/1Q27, (2) the

Chao Wang

+886(2)2730-4195 | kuan-chao.wang@gs.com
Goldman Sachs (Asia) L.L.C., Taipei Branch

Allen Chang

+852-2978-2930 | allen.k.chang@gs.com
Goldman Sachs (Asia) L.L.C.

Al Wang

+886(2)2730-4081 | al.wang@gs.com
Goldman Sachs (Asia) L.L.C., Taipei Branch

Goldman Sachs does and seeks to do business with companies covered in its research reports. As a result, investors should be aware that the firm may have a conflict of interest that could affect the objectivity of this report. Investors should consider this report as only a single factor in making their investment decision. For Reg AC certification and other important disclosures, see the Disclosure Appendix, or go to www.gs.com/research/hedge.html. Analysts employed by non-US affiliates are not registered/qualified as research analysts with FINRA in the U.S.

ramp-up of D2C satellites from 2H27, with PCB area consumption to further increase by 40-50%, and (3) the ramp-up of AI satellites from 2H27, with PCB dollar content value that is similar to LEO satellites, but at a significantly larger annual deployment volume of several tens of thousands of units (vs. 3-4K LEO satellites launched by the key LEO customer in 2025).

(6) The company mentioned that it is seeing some impact from material shortages, but did mention the supply conditions for low-end CCL are improving, as supply has been primarily constrained by E-glass. However, management mentioned there are no supply issues for high-end CCL, while supply is well-supported by key CCL suppliers. In terms of rising material costs, the company said it can pass through cost increase across nearly all products, except for SLP products for the key smartphone customer.

(7) The company said it is seeing multiple growth opportunities for its data center PCB business, including (1) strong demand for high-end PCB for AI satellites from its key LEO customers (as mentioned in bullet point 4), (2) PCB for AI servers from the key smartphone customer, which is expected to ramp-up from 4Q26, (3) additional contributions from China customers with PCB demand across switches and AI servers, and (4) an ongoing project with a small-scale ASIC client.

PCB/CCL sector read-across

We retain our positive view on the optical transceivers' mSAP demand outlook, while our supply chain checks recently suggested mSAP as a key bottleneck for the 1.6T optical transceivers supply chain now, whose product pricing has gone up by 30%+ YTD with good profitability (our checks and Compeq's comments suggest the GM for 1.6T optical transceivers mSAP are generally 40-45%+). We believe ZDT has the best opportunity to gain more market share in this high-growth and profitable market (we expect TAM for 1.6T+ optical transceivers mSAP to go up from US\$0.8bn in 2026 to US\$2.9bn in 2028, implying 94% CAGR), considering it has the most aggressive mSAP capacity expansion plan (we expect ZDT to expand mSAP capacity by 5x in the next two years) and the company's best-in-class BT/mSAP production technology, which could become the company's key revenue driver (we expect mSAP to account for ~8%/10%+ of ZDT revenue in 2027/28 from ~2% in 2026) and margin driver (mSAP GM should be 40-45%+ vs. ZDT 1H26 GM at only ~22+%).

We also expect CCL suppliers, such as EMC and TUC, to benefit from the strong optical transceiver demand, and we expect both companies to gain more share with more CCL capacity to be ready in 2027/28 (see [here](#) and [here](#)).

Overall, we retain our positive view on ZDT's business outlook not only on mSAP products, but also seeing a significant margin expansion opportunity from the high-end AI PCB and ABF substrate products, where we also see a strong demand outlook (see our AI PCB industry report [here](#) and our ABF substrate industry report [here](#)).

We maintain our Buy ratings on ZDT, EMC and TUC with unchanged 12m TPs of NT\$925/NT\$10,200/NT\$3,010.

Investment Thesis & PT methodology and risks

Investment Thesis - Zhen Ding Technology

Zhen Ding Technology (ZDT) is the global leading PCB supplier (top 1 in market share in sales) with 7-10% market share, and started its ABF substrate business from 2023 in Shenzhen, China; it has the highest ABF revenue exposure to the China market compared to all other ABF substrate suppliers. We believe ZDT will hold more than 40% ABF substrate market share in China by 2027E (up from 31% in 2024) and benefit the most from the strong and growing China ABF substrate market (53% 2025E-27E CAGR) due to the increasing in-sourcing ratio for the AI computing ASIC/GPU market in China. We also expect the increasing foldable smartphone penetration rate in the long term to drive ZDT's FPC TAM, and account for 50% of the company's total revenue in 2027E. We see further revenue upside potential from foldable phones. We are Buy rated mainly due to our positive view on the company's China ABF substrate business opportunity, which should not only drive ZDT's revenue growth, but also improve its OPM level in the long term. We see upside for the stock price considering the company's solid growth opportunity, and our TP implied P/E multiple is still lower than the company's past 10+ year P/E average, which we believe is fair considering the potential upside from China ABF substrate market and increasing foldable phone penetration rate.

Valuation: Our 12-month TP of NT\$925 is based on a SOTP valuation methodology, including (1) 3.5x blended 2028E P/B for ABF & BT substrate, (2) 20x 2028E P/E for high-growth AI server/switch related PCB (in line with industry average), and (3) 8x 2028E P/E for low-growth consumer electronics PCB (in-line with industry downcycle average valuation).

Key downside risks include: (1) weaker-than-expected iPhone demand or slower content upgrade, (2) delay in new substrate capacity or slower-than-expected yield rate ramp up progress, and (3) fiercer competition in China ABF market.

Investment Thesis - Elite Material

EMC, a key supplier of high-end HDI material (70%+ market share) and SLP material (90%+ market share), has been focusing on the high-speed switch/server CCL market for 3-5+ years. We are positive on its market share trajectory in the server industry and expect EMC to increase its share from <10% in the Purley generation to 15-20%/20+% in the Whitley/Eagle Stream platforms. EMC is also proactively expanding its share in the switch market (40%/30%+ in 2024/23 vs. <5% before 2019), with an expanding AI customer base (only one 400G customer in 3Q20, to being one of the largest suppliers in 2023), which should continue to benefit the company's top/bottom line growth over time. We believe EMC's leading position in the AI server CCL market should drive strong revenue growth and GM/OPM expansion in the long term, given that it is the major supplier for all Nvidia/Google/AWS AI server projects, and we expect it to maintain its leadership position. Considering its leading position in the high-end CCL industry, solid production skills and earnings growth outlook, we are Buy rated on the stock. With the shares trading at a 2026E P/E below AI server component suppliers and supported by a solid growth outlook, we view valuation as attractive.

Key downside risks include: (1) RCC to replace all high-end smartphone HDI design; (2) rising trade tensions, which could lead to weaker smartphone and server shipments; and

(3) rising competition from mainland China peers.

Price Target Risks and Methodology – Elite Material

Valuation methodology: Our 12m TP of NT\$10,200 is based on 27x 2H27-1H28E P/E (in line with the peak P/E multiple for AI component suppliers in the past 3 years).

Key downside risks: (1) RCC to replace all high-end smartphone HDI design; (2) rising trade tensions, which could lead to weaker smartphone and server shipments; and (3) rising competition from mainland China peers.

Investment Thesis – Taiwan Union Technology Corp.

TUC is a key high-end global CCL supplier that focuses on M7+ grade high-speed CCL (key applications include high-end switches (100G+) and AI server materials), with 20%+ market share in the past two years. The company continues to expect its unit market share in the server industry to increase from 15% in Whitley and Purley processors to higher in the Eagle Stream generation, and plans to launch two types of products for low-end and high-end customers (T2A and T2C) to gain market share. Also, strong growth in 400G/800G switch shipments and new 800G switch shipments, scheduled for launch in 2025-27, should continue to be a key demand driver for TUC, in our view. Additionally, the company is working on high-end AI projects with CSP and enterprise players, which we believe should drive revenue momentum in the long term. We view the stock as undervalued vs. Taiwan CCL peers on a P/E basis and rate it Buy.

Key downside risks include: (1) slower-than-expected share gains in the low-loss CCL segment; (2) rising trade tensions, which could lead to weaker server and switch shipments globally; and (3) rising competition from mainland China peers.

Price Target Risks and Methodology – Taiwan Union Technology Corp.

Valuation methodology: Our 12m TP of NT\$3,010 is based on a 22x 2027E P/E (+2x STDV higher than the past 3-year industry average P/E multiple).

Key downside risks: (1) slower-than-expected share gains in the low-loss CCL segment; (2) rising trade tensions, which could lead to weaker server and switch shipments globally; and (3) rising competition from mainland China peers.

Disclosure Appendix

Reg AC

We, Chao Wang, Allen Chang and Al Wang, hereby certify that all of the views expressed in this report accurately reflect our personal views about the subject company or companies and its or their securities. We also certify that no part of our compensation was, is or will be, directly or indirectly, related to the specific recommendations or views expressed in this report.

Unless otherwise stated, the individuals listed on the cover page of this report are analysts in Goldman Sachs' Global Investment Research division.

Contributing Authors: Chao Wang Goldman Sachs (Asia) L.L.C., Taipei Branch, Allen Chang Goldman Sachs (Asia) L.L.C., Al Wang Goldman Sachs (Asia) L.L.C., Taipei Branch.

Unless otherwise stated, the individuals listed in the Contributing Authors disclosure of this report are analysts in Goldman Sachs' Global Investment Research division.

GS Factor Profile

The Goldman Sachs Factor Profile provides investment context for a stock by comparing key attributes to the market (i.e. our universe of rated stocks) and its sector peers. The four key attributes depicted are: Growth, Financial Returns, Multiple (e.g. valuation) and Integrated (a composite of Growth, Financial Returns and Multiple). Growth, Financial Returns and Multiple are calculated by using normalized ranks for specific metrics for each stock. The normalized ranks for the metrics are then averaged and converted into percentiles for the relevant attribute. The precise calculation of each metric may vary depending on the fiscal year, industry and region, but the standard approach is as follows:

Growth is based on a stock's forward-looking sales growth, EBITDA growth and EPS growth (for financial stocks, only EPS and sales growth), with a higher percentile indicating a higher growth company. **Financial Returns** is based on a stock's forward-looking ROE, ROCE and CROCI (for financial stocks, only ROE), with a higher percentile indicating a company with higher financial returns. **Multiple** is based on a stock's forward-looking P/E, P/B, price/dividend (P/D), EV/EBITDA, EV/FCF and EV/Debt Adjusted Cash Flow (DACF) (for financial stocks, only P/E, P/B and P/D), with a higher percentile indicating a stock trading at a higher multiple. The **Integrated** percentile is calculated as the average of the Growth percentile, Financial Returns percentile and (100% - Multiple percentile).

Financial Returns and Multiple use the Goldman Sachs analyst forecasts at the fiscal year-end at least three quarters in the future. Growth uses inputs for the fiscal year at least seven quarters in the future compared with the year at least three quarters in the future (on a per-share basis for all metrics).

For a more detailed description of how we calculate the GS Factor Profile, please contact your GS representative.

M&A Rank

Across our global coverage, we examine stocks using an M&A framework, considering both qualitative factors and quantitative factors (which may vary across sectors and regions) to incorporate the potential that certain companies could be acquired. We then assign a M&A rank as a means of scoring companies under our rated coverage from 1 to 3, with 1 representing high (30%-50%) probability of the company becoming an acquisition target, 2 representing medium (15%-30%) probability and 3 representing low (0%-15%) probability. For companies ranked 1 or 2, in line with our standard departmental guidelines we incorporate an M&A component into our target price. M&A rank of 3 is considered immaterial and therefore does not factor into our price target, and may or may not be discussed in research.

Quantum

Quantum is Goldman Sachs' proprietary database providing access to detailed financial statement histories, forecasts and ratios. It can be used for in-depth analysis of a single company, or to make comparisons between companies in different sectors and markets.

Disclosures

The rating(s) for Elite Material, Taiwan Union Technology Corp. and Zhen Ding Technology Holding is/are relative to the other companies in its/their coverage universe: AWSC, Airtac International Group, Co-Tech Development Corp., Delta Electronics, Elite Material, GCE, Hiwin Corp., ITEQ Corp, Kinsus, Lotes, NYPCB, Taiwan Union Technology Corp., Unimicron Technology, Win Semiconductors Corp., Yageo Corp., Zhen Ding Technology Holding

Company-specific regulatory disclosures

The following disclosures relate to relationships between The Goldman Sachs Group, Inc. (with its affiliates, "Goldman Sachs") and companies covered by Goldman Sachs Global Investment Research and referred to in this research.

Goldman Sachs beneficially owned 1% or more of common equity (excluding positions managed by affiliates and business units not required to be aggregated under US securities law) as of the second most recent month end: Compeq Manufacturing (NT\$249.50) and Taiwan Union Technology Corp. (NT\$1,365.00)

Goldman Sachs expects to receive or intends to seek compensation for investment banking services in the next 3 months: Elite Material (NT\$5,415.00) and Zhen Ding Technology Holding (NT\$463.00)

Goldman Sachs had an investment banking services client relationship during the past 12 months with: Zhen Ding Technology Holding (NT\$463.00)

Distribution of ratings/investment banking relationships

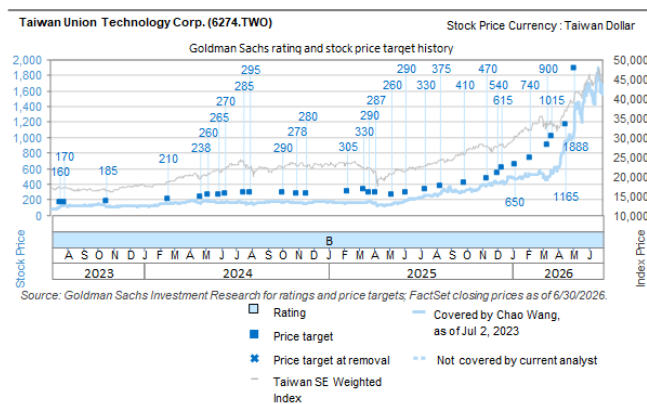
Goldman Sachs Investment Research global Equity coverage universe

	Rating Distribution				Investment Banking Relationships		
	Buy	Hold	Sell		Buy	Hold	Sell
Global	50%	34%	16%		65%	59%	43%

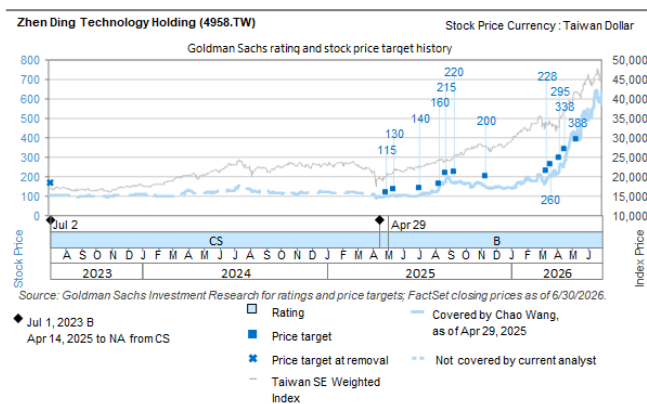
As of July 1, 2026, Goldman Sachs Global Investment Research had investment ratings on 3,104 equity securities. Goldman Sachs assigns stocks as Buys and Sells on various regional Investment Lists; stocks not so assigned are deemed Neutral. Such assignments equate to Buy, Hold and Sell for the purposes of the above disclosure required by the FINRA Rules. See 'Ratings, Coverage universe and related definitions' below. The Investment Banking Relationships chart reflects the percentage of subject companies within each rating category for whom Goldman Sachs has provided investment

banking services within the previous twelve months.

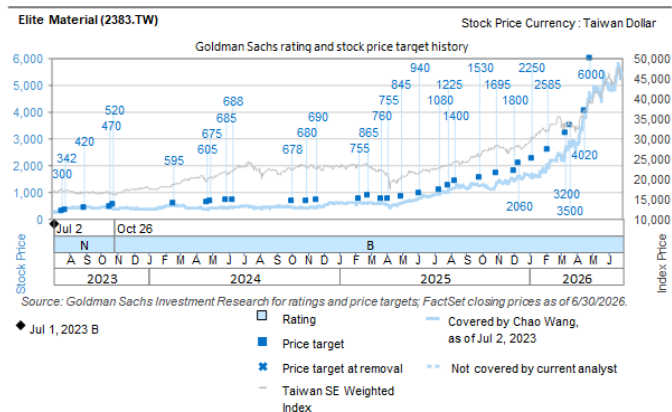
Price target and rating history chart(s)



The price targets shown should be considered in the context of all prior published Goldman Sachs research, which may or may not have included price targets, as well as developments relating to the company, its industry and financial markets.



The price targets shown should be considered in the context of all prior published Goldman Sachs research, which may or may not have included price targets, as well as developments relating to the company, its industry and financial markets.



The price targets shown should be considered in the context of all prior published Goldman Sachs research, which may or may not have included price targets, as well as developments relating to the company, its industry and financial markets.

Target price history table(s)

Taiwan Union Technology Corp. (6274.TWO)

Date of report	Target price (NT\$)	Closing price (NT\$)
29-Jul-26	3,010.00	1,020.00
17-Jul-26	2,860.00	1,170.00
06-May-26	1,888.00	1,315.00
17-Apr-26	1,165.00	946.00
20-Mar-26	1,015.00	554.00
12-Mar-26	900.00	452.50
05-Feb-26	740.00	520.00
06-Jan-26	650.00	476.00
12-Dec-25	615.00	443.00
04-Dec-25	540.00	411.00
12-Nov-25	470.00	407.00
29-Sep-25	410.00	298.00
13-Aug-25	375.00	282.50
15-Jul-25	330.00	264.00
05-Jun-25	290.00	188.50
07-May-25	260.00	142.50
06-Apr-25	287.00	159.00
24-Mar-25	290.00	176.50
12-Mar-25	330.00	160.50
09-Feb-25	305.00	163.50

Elite Material (2383.TW)

Date of report	Target price (NT\$)	Closing price (NT\$)
29-Jul-26	10,200.00	4,100.00
17-Jul-26	9,500.00	4,495.00
29-Apr-26	6,000.00	4,440.00
17-Apr-26	4,020.00	3,800.00
20-Mar-26	3,500.00	2,810.00
12-Mar-26	3,200.00	2,600.00
05-Feb-26	2,585.00	1,950.00
06-Jan-26	2,250.00	1,695.00
12-Dec-25	2,060.00	1,605.00
04-Dec-25	1,800.00	1,425.00
30-Oct-25	1,695.00	1,275.00
29-Sep-25	1,530.00	1,225.00
13-Aug-25	1,400.00	1,200.00
30-Jul-25	1,225.00	1,005.00
15-Jul-25	1,080.00	970.00
05-Jun-25	940.00	789.00
30-Apr-25	845.00	556.00
06-Apr-25	755.00	527.00
24-Mar-25	760.00	604.00
25-Feb-25	865.00	576.00

Taiwan Union Technology Corp. (6274.TWO)

Date of report	Target price (NT\$)	Closing price (NT\$)
20-Nov-24	280.00	155.00
30-Oct-24	278.00	162.00
03-Oct-24	290.00	175.00
31-Jul-24	295.00	160.00
17-Jul-24	285.00	168.00
11-Jun-24	270.00	164.50
29-May-24	265.00	167.00
08-May-24	260.00	185.50
23-Apr-24	238.00	164.50
18-Feb-24	210.00	156.50
20-Oct-23	185.00	111.00

Elite Material (2383.TW)

Date of report	Target price (NT\$)	Closing price (NT\$)
09-Feb-25	755.00	615.00
20-Nov-24	690.00	447.50
30-Oct-24	680.00	419.50
03-Oct-24	678.00	442.00
11-Jun-24	688.00	422.50
29-May-24	685.00	427.00
30-Apr-24	675.00	412.00
23-Apr-24	605.00	376.00
18-Feb-24	595.00	534.00
26-Oct-23	520.00	388.50
20-Oct-23	470.00	403.00

Zhen Ding Technology Holding (4958.TW)

Date of report	Target price (NT\$)	Closing price (NT\$)
11-Aug-26	925.00	490.00
12-Jul-26	824.00	588.00
12-May-26	388.00	407.00
17-Apr-26	338.00	282.00
08-Apr-26	295.00	247.50
20-Mar-26	260.00	209.50
13-Mar-26	228.00	188.50
12-Nov-25	200.00	144.50
10-Sep-25	220.00	168.50
25-Aug-25	215.00	180.50
12-Aug-25	160.00	147.00
04-Jul-25	140.00	103.00
13-May-25	130.00	104.50
29-Apr-25	115.00	97.10

Price targets shown in table(s) are unadjusted for corporate actions.

Regulatory disclosures**Disclosures required by United States laws and regulations**

See company-specific regulatory disclosures above for any of the following disclosures required as to companies referred to in this report: manager or co-manager in a pending transaction; 1% or other ownership; compensation for certain services; types of client relationships; managed/co-managed public offerings in prior periods; directorships; registered market maker. Goldman Sachs trades or may trade as a principal in debt securities (or in related derivatives) of issuers discussed in this report.

The following are additional required disclosures: **Ownership and material conflicts of interest:** Goldman Sachs policy prohibits its analysts, professionals reporting to analysts and members of their households from owning securities of any company in the analyst's area of coverage. **Analyst compensation:** Analysts are paid in part based on the profitability of Goldman Sachs, which includes investment banking revenues. **Analyst as officer or director:** Goldman Sachs policy generally prohibits its analysts, persons reporting to analysts or members of their households from serving as an officer, director or advisor of any company in the analyst's area of coverage. **Non-U.S. Analysts:** Non-U.S. analysts may not be associated persons of Goldman Sachs & Co. LLC and therefore may not be subject to FINRA Rule 2241 or FINRA Rule 2242 restrictions on communications with a subject company, public appearances and trading in securities covered by the analysts.

Additional disclosures required under the laws and regulations of jurisdictions other than the United States

The following disclosures are those required by the jurisdiction indicated, except to the extent already made above pursuant to United States laws and regulations. **Australia:** Goldman Sachs Australia Pty Ltd and its affiliates are not authorised deposit-taking institutions (as that term is defined in the Banking Act 1959 (Cth)) in Australia and do not provide banking services, nor carry on a banking business, in Australia. This research, and any access to it, is intended only for "wholesale clients" within the meaning of the Australian Corporations Act, unless otherwise agreed by Goldman Sachs. In producing research reports, members of Global Investment Research of Goldman Sachs Australia may attend site visits and other meetings hosted by the companies and other entities which are the subject of its research reports. In some instances the costs of such site visits or meetings may be met in part or in whole by the issuers concerned if Goldman Sachs Australia considers it is appropriate and reasonable in the specific circumstances relating to the site visit or meeting. To the extent that the contents of this document contains any financial product advice, it is general advice only and has been prepared by Goldman Sachs without taking into account a client's objectives, financial situation or needs. A client should, before acting on any such advice, consider the appropriateness of the advice having regard to the client's own objectives, financial situation and needs. A copy of certain Goldman Sachs Australia and New Zealand disclosure of interests and a copy of Goldman Sachs' Australian Sell-Side Research Independence Policy Statement are available at: <https://www.goldmansachs.com/disclosures/australia-new-zealand/index.html>. **Brazil:** Disclosure information in relation to CVM Resolution n. 20 is available at <https://www.gs.com/worldwide/brazil/area/gir/index.html>. Where applicable, the Brazil-registered analyst primarily responsible for the content of this research report, as defined in Article 20 of CVM Resolution n. 20, is the first author named at the beginning of this report, unless indicated otherwise at the end of the text. **Canada:** This information is being provided to you for information purposes only and is not, and under no circumstances should be construed as, an advertisement, offering or solicitation by Goldman Sachs & Co. LLC for purchasers of securities in Canada to trade in any Canadian security. Goldman Sachs & Co. LLC is not registered as a dealer in any jurisdiction in Canada under applicable Canadian securities laws and generally is not permitted to trade in Canadian securities and may be prohibited from selling certain securities and products in certain jurisdictions in Canada. If you wish to trade in any Canadian securities or other products in Canada please contact Goldman

Sachs Canada Inc., an affiliate of The Goldman Sachs Group Inc., or another registered Canadian dealer. **Hong Kong:** Further information on the securities of covered companies referred to in this research may be obtained on request from Goldman Sachs (Asia) L.L.C. **India:** Further information on the subject company or companies referred to in this research may be obtained from Goldman Sachs (India) Securities Private Limited, Research Analyst – SEBI Registration Number INH000001493, 10th Floor, Ascent-Worli, Sudam Kalu Ahire Marg, Worli, Mumbai-400 025, India, Corporate Identity Number U74140MH2006FTC160634, Phone +91 22 6616 9000, Fax +91 22 6616 9001. Goldman Sachs may beneficially own 1% or more of the securities (as such term is defined in clause 2 (h) the Indian Securities Contracts (Regulation) Act, 1956) of the subject company or companies referred to in this research report. Registration granted by SEBI and certification from NISM in no way guarantee performance of the intermediary or provide any assurance of returns to investors. Goldman Sachs (India) Securities Private Limited compliance officer and investor grievance contact details, a copy of the annual compliance audit report and other relevant information and disclosures can be found at this link: <https://www.goldmansachs.com/worldwide/india/research-analyst>. **Japan:** See below. **Korea:** This research, and any access to it, is intended only for “professional investors” within the meaning of the Financial Services and Capital Markets Act, unless otherwise agreed by Goldman Sachs. Further information on the subject company or companies referred to in this research may be obtained from Goldman Sachs (Asia) L.L.C., Seoul Branch. **New Zealand:** Goldman Sachs New Zealand Limited and its affiliates are neither “registered banks” nor “deposit takers” (as defined in the Reserve Bank of New Zealand Act 1989) in New Zealand. This research, and any access to it, is intended for “wholesale clients” (as defined in the Financial Advisers Act 2008) unless otherwise agreed by Goldman Sachs. A copy of certain Goldman Sachs Australia and New Zealand disclosure of interests is available at: <https://www.goldmansachs.com/disclosures/australia-new-zealand/index.html>. **Russia:** Research reports distributed in the Russian Federation are not advertising as defined in the Russian legislation, but are information and analysis not having product promotion as their main purpose and do not provide appraisal within the meaning of the Russian legislation on appraisal activity. Research reports do not constitute a personalized investment recommendation as defined in Russian laws and regulations, are not addressed to a specific client, and are prepared without analyzing the financial circumstances, investment profiles or risk profiles of clients. Goldman Sachs assumes no responsibility for any investment decisions that may be taken by a client or any other person based on this research report. **Singapore:** Goldman Sachs (Singapore) Pte. (Company Number: 198602165W), which is regulated by the Monetary Authority of Singapore, accepts legal responsibility for this research, and should be contacted with respect to any matters arising from, or in connection with, this research. **Taiwan:** This material is for reference only and must not be reprinted without permission. Investors should carefully consider their own investment risk. Investment results are the responsibility of the individual investor. **United Kingdom:** Persons who would be categorized as retail clients in the United Kingdom, as such term is defined in the rules of the Financial Conduct Authority, should read this research in conjunction with prior Goldman Sachs research on the covered companies referred to herein and should refer to the risk warnings that have been sent to them by Goldman Sachs International. A copy of these risks warnings, and a glossary of certain financial terms used in this report, are available from Goldman Sachs International on request.

European Union and United Kingdom: Disclosure information in relation to Article 6 (2) of the European Commission Delegated Regulation (EU) (2016/958) supplementing Regulation (EU) No 596/2014 of the European Parliament and of the Council (including as that Delegated Regulation is implemented into United Kingdom domestic law and regulation following the United Kingdom’s departure from the European Union and the European Economic Area) with regard to regulatory technical standards for the technical arrangements for objective presentation of investment recommendations or other information recommending or suggesting an investment strategy and for disclosure of particular interests or indications of conflicts of interest is available at <https://www.gs.com/disclosures/europeanpolicy.html> which states the European Policy for Managing Conflicts of Interest in Connection with Investment Research.

Japan: Goldman Sachs Japan Co., Ltd. is a Financial Instrument Dealer registered with the Kanto Financial Bureau under registration number Kinsho 69, and a member of Japan Securities Dealers Association, Financial Futures Association of Japan Type II Financial Instruments Firms Association, and Investment Management Association of Japan. Sales and purchase of equities are subject to commission pre-determined with clients plus consumption tax. See company-specific disclosures as to any applicable disclosures required by Japanese stock exchanges, the Japanese Securities Dealers Association or the Japanese Securities Finance Company.

Ratings, coverage universe and related definitions

Buy (B), Neutral (N), Sell (S) Analysts recommend stocks as Buys or Sells for inclusion on various regional Investment Lists. Being assigned a Buy or Sell on an Investment List is determined by a stock’s total return potential relative to its coverage universe. Any stock not assigned as a Buy or a Sell on an Investment List with an active rating (i.e., a stock that is not Rating Suspended, Not Rated, Early-Stage Biotech, Coverage Suspended or Not Covered), is deemed Neutral. Each region manages Regional Conviction Lists, which are selected from Buy rated stocks on the respective region’s Investment Lists and represent investment recommendations focused on the size of the total return potential and/or the likelihood of the realization of the return across their respective areas of coverage. The addition or removal of stocks from such Conviction Lists are managed by the Investment Review Committee or other designated committee in each respective region and do not represent a change in the analysts’ investment rating for such stocks.

Total return potential represents the upside or downside differential between the current share price and the price target, including all paid or anticipated dividends, expected during the time horizon associated with the price target. Price targets are required for all covered stocks. The total return potential, price target and associated time horizon are stated in each report adding or reiterating an Investment List membership.

Coverage Universe: A list of all stocks in each coverage universe is available by primary analyst, stock and coverage universe at <https://www.gs.com/research/hedge.html>.

Not Rated (NR). The investment rating, target price and earnings estimates (where relevant) are removed pursuant to Goldman Sachs policy when Goldman Sachs is acting in an advisory capacity in a merger or in a strategic transaction involving this company, when there are legal, regulatory or policy constraints due to Goldman Sachs’ involvement in a transaction, and in certain other circumstances. **Early-Stage Biotech (ES).** An investment rating and a target price are not assigned pursuant to Goldman Sachs policy when this company has neither a drug, treatment or medical device that has passed a Phase II clinical trial nor a license to distribute a post-Phase II drug, treatment or medical device. **Rating Suspended (RS).** Goldman Sachs Research has suspended the investment rating and price target for this stock, because there is not a sufficient fundamental basis for determining an investment rating or target price. The previous investment rating and target price, if any, are no longer in effect for this stock and should not be relied upon. **Coverage Suspended (CS).** Goldman Sachs has suspended coverage of this company. **Not Covered (NC).** Goldman Sachs does not cover this company.

Global product; distributing entities

Goldman Sachs Global Investment Research produces and distributes research products for clients of Goldman Sachs on a global basis. Analysts based in Goldman Sachs offices around the world produce research on industries and companies, and research on macroeconomics, currencies, commodities and portfolio strategy. This research is disseminated in Australia by Goldman Sachs Australia Pty Ltd (ABN 21 006 797 897); in Brazil by Goldman Sachs do Brasil Corretora de Títulos e Valores Mobiliários S.A.; Public Communication Channel Goldman Sachs Brazil: 0800 727 5764 and / or contatogoldmanbrasil@gs.com. Available Weekdays (except holidays), from 9am to 6pm. Canal de Comunicação com o Público Goldman Sachs Brasil: 0800 727 5764 e/ou contatogoldmanbrasil@gs.com. Horário de funcionamento: segunda-feira à sexta-feira (exceto feriados), das 9h às 18h; in Canada by Goldman Sachs & Co. LLC; in Hong Kong by Goldman Sachs (Asia) L.L.C.; in India by Goldman Sachs (India) Securities Private Ltd.; in Japan by Goldman Sachs Japan Co., Ltd.; in the Republic of Korea by Goldman Sachs (Asia) L.L.C., Seoul Branch; in New Zealand by Goldman Sachs New Zealand Limited; in Russia by OOO Goldman Sachs; in Singapore by Goldman Sachs (Singapore) Pte. (Company Number: 198602165W); and in the United States of America by Goldman Sachs & Co. LLC. Goldman Sachs International has approved this research in connection with its distribution in the United Kingdom.

Goldman Sachs International ("GSI"), authorised by the Prudential Regulation Authority ("PRA") and regulated by the Financial Conduct Authority ("FCA") and the PRA, has approved this research in connection with its distribution in the United Kingdom.

European Economic Area: Goldman Sachs Bank Europe SE ("GSBE") is a credit institution incorporated in Germany and, within the Single Supervisory Mechanism, subject to direct prudential supervision by the European Central Bank and in other respects supervised by German Federal Financial Supervisory Authority (Bundesanstalt für Finanzdienstleistungsaufsicht, BaFin) and Deutsche Bundesbank and disseminates research within the European Economic Area.

General disclosures

This research is for our clients only. Other than disclosures relating to Goldman Sachs, this research is based on current public information that we consider reliable, but we do not represent it is accurate or complete, and it should not be relied on as such. The information, opinions, estimates and forecasts contained herein are as of the date hereof and are subject to change without prior notification. We seek to update our research as appropriate, but various regulations may prevent us from doing so. Other than certain industry reports published on a periodic basis, the large majority of reports are published at irregular intervals as appropriate in the analyst's judgment.

Goldman Sachs conducts a global full-service, integrated investment banking, investment management, and brokerage business. We have investment banking and other business relationships with a substantial percentage of the companies covered by Global Investment Research. Goldman Sachs & Co. LLC, the United States broker dealer, is a member of SIPC (<https://www.sipc.org>).

Our salespeople, traders, and other professionals may provide oral or written market commentary or trading strategies to our clients and principal trading desks that reflect opinions that are contrary to the opinions expressed in this research. Our asset management area, principal trading desks and investing businesses may make investment decisions that are inconsistent with the recommendations or views expressed in this research.

The analysts named in this report may have from time to time discussed with our clients, including Goldman Sachs salespersons and traders, or may discuss in this report, trading strategies that reference catalysts or events that may have a near-term impact on the market price of the equity securities discussed in this report, which impact may be directionally counter to the analyst's published price target expectations for such stocks. Any such trading strategies are distinct from and do not affect the analyst's fundamental equity rating for such stocks, which rating reflects a stock's return potential relative to its coverage universe as described herein.

We and our affiliates, officers, directors, and employees will from time to time have long or short positions in, act as principal in, make a market or act as liquidity provider in, and buy or sell, any securities, derivatives, or other financial instruments referred to in this research (including any financial instruments issued by the companies referred to herein), unless otherwise prohibited by regulation or Goldman Sachs policy.

The views attributed to third party presenters at Goldman Sachs arranged conferences, including individuals from other parts of Goldman Sachs, do not necessarily reflect those of Global Investment Research and are not an official view of Goldman Sachs.

Any third party referenced herein, including any salespeople, traders and other professionals or members of their household, may have positions in the products mentioned that are inconsistent with the views expressed by analysts named in this report.

This research is not an offer to sell or the solicitation of an offer to buy any security in any jurisdiction where such an offer or solicitation would be illegal. It does not constitute a personal recommendation or take into account the particular investment objectives, financial situations, or needs of individual clients. Clients should consider whether any advice or recommendation in this research is suitable for their particular circumstances and, if appropriate, seek professional advice, including tax advice. The price and value of investments referred to in this research and the income from them may fluctuate. Past performance is not a guide to future performance, future returns are not guaranteed, and a loss of original capital may occur. Fluctuations in exchange rates could have adverse effects on the value or price of, or income derived from, certain investments.

Certain transactions, including those involving futures, options, and other derivatives, give rise to substantial risk and are not suitable for all investors. Investors should review current options and futures disclosure documents which are available from Goldman Sachs sales representatives or at <https://www.theocc.com/about/publications/character-risks.jsp> and https://www.goldmansachs.com/disclosures/cftc_fcm_disclosures. Transaction costs may be significant in option strategies calling for multiple purchase and sales of options such as spreads. Supporting documentation will be supplied upon request.

Differing Levels of Service provided by Global Investment Research: The level and types of services provided to you by Goldman Sachs Global Investment Research may vary as compared to that provided to internal and other external clients of GS, depending on various factors including your individual preferences as to the frequency and manner of receiving communication, your risk profile and investment focus and perspective (e.g., marketwide, sector specific, long term, short term), the size and scope of your overall client relationship with GS, and legal and regulatory constraints. As an example, certain clients may request to receive notifications when research on specific securities is published, and certain clients may request that specific data underlying analysts' fundamental analysis available on our internal client websites be delivered to them electronically through data feeds or otherwise. No change to an analyst's fundamental research views (e.g., ratings, price targets, or material changes to earnings estimates for equity securities), will be communicated to any client prior to inclusion of such information in a research report broadly disseminated through electronic publication to our internal client websites or through other means, as necessary, to all clients who are entitled to receive such reports.

All research reports are disseminated and available to all clients simultaneously through electronic publication to our internal client websites. Not all research content is redistributed to our clients or available to third-party aggregators, nor is Goldman Sachs responsible for the redistribution of our research by third party aggregators. For research, models or other data related to one or more securities, markets or asset classes (including related services) that may be available to you, please contact your GS representative or go to <https://research.gs.com>.

Disclosure information is also available at <https://www.gs.com/research/hedge.html> or from Research Compliance, 200 West Street, New York, NY 10282.

© 2026 Goldman Sachs.

You are permitted to store, display, analyze, modify, reformat, and print the information made available to you via this service only for your own use. You may not resell or reverse engineer this information to calculate or develop any index for disclosure and/or marketing or create any other derivative works or commercial product(s), data or offering(s) without the express written consent of Goldman Sachs. You are not permitted to publish, transmit, or otherwise reproduce this information, in whole or in part, in any format to any third party without the express written consent of Goldman Sachs. This foregoing restriction includes, without limitation, using, extracting, downloading or retrieving this information, in whole or in part, to train or finetune a machine learning or artificial intelligence system, or to provide or reproduce this information, in whole or in part, as a prompt or input to any such system.